

Vol 14 Chapter 6 — Bell Sub-Tsirelson as Information Bound

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 6 — Bell Sub-Tsirelson as Information Bound

[FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.] *(Grace's R143 G2 bracket, moved out of the YAML front matter — where it broke the PDF build — into the body, verbatim; Keeper 2026-09-13.)*

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

This chapter's prediction FIRED — and it had fired before the chapter was written (2026-09-11, Keeper, K1893; Grace Round 142 G3; falsifier register E4). The sub-Tsirelson ceiling this chapter presents — $S_{BST} = \sqrt{126/16} = 2.80624$, a deficit of exactly $1/2^{N_c} = 1/8$ in S^2 below Tsirelson's $2\sqrt{2}$ — was refuted by Poh, Joshi, Ceré, Cabello and Kurtsiefer, *Phys. Rev. Lett.* 115, 180408 (October 2015; arXiv:1506.01865), who measured $S = 2.82759 \pm 0.00051$, a distance 0.00084 ± 0.00051 from Tsirelson's bound and 41.9σ above ours. The prediction was registered in May 2026 with a six-to-twelve-month window for an experiment that had been done ten years earlier at higher precision than proposed. What died: the ceiling as a prediction, the Bell-apparatus proposal, and this chapter as a standing result. What survives: the integer identity, and $\text{Tr}(B^2) = 126/16$ as a fact about an operator on $H^2(D_{IV}^5)$ with no laboratory claim. The Tsirelson bound itself is untouched — the quantum mechanics recovered on the Hardy space is standard QM, whose bound is $2\sqrt{2}$; the fired number was a coding reading laid over it. The chapter is kept below as the May record, with the same ceremony as a win: it could have failed, and this is where it did. Spine Lecture 3 carries the loss in the derived core.

Level 1 — one sentence

BST predicts Bell-CHSH correlations below Tsirelson's quantum bound by exactly $1/2^{N_c} = 1/8$: $S_{BST}^2 = S_{Tsirelson}^2 - 1/8 = 8 - 1/8 = 63/8$, equivalently $S_{BST} \leq \sqrt{63/8} \approx 2.8062$ vs Tsirelson's $2\sqrt{2} \approx 2.8284$, representing the substrate's finite operational power per commitment cycle (SCMP principle, T2469, Vol 5 Ch 8) — a testable falsifier at $\sim 0.78\%$ deviation distinguishing BST from standard QM. [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]

Level 2 — graduate-physicist precision

6.1 CHSH inequality

Clauser-Horne-Shimony-Holt 1969: classical local-hidden-variable theories satisfy

$$|S| = |E(a, b) + E(a, b') + E(a', b) - E(a', b')| \leq 2$$

with correlations $E(a, b) = \langle A(a)B(b) \rangle$.

6.2 Tsirelson bound

Tsirelson 1980: quantum mechanics allows $|S| \leq 2\sqrt{2} \approx 2.8284$.

Achieved by maximally entangled Bell pair with appropriate measurement angles.

6.3 BST sub-Tsirelson prediction [FIRED 2026-09-13 — register E4; see the bracket below]

T2469 (Casey's SCMP principle): BST substrate has finite operational power per commitment cycle, yielding strict reduction below Tsirelson: [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]

$$S_{BST}^2 = S_{Tsirelson}^2 - 1/2^{N_c} = 8 - 1/8 = 63/8$$

$$S_{BST} \leq 2 \cdot \sqrt{63/32} \approx 2.8062$$

Deviation from Tsirelson: $(2\sqrt{2} - 2.8062)/(2\sqrt{2}) \approx 0.78\%$.

6.4 Information-theoretic reading

The substrate commits to one outcome per Koons tick at finite channel capacity $g = 7$ bits/tick (Vol 14 Ch 3).

Quantum mechanics in its idealized formulation assumes infinite-precision commitment (Tsirelson saturation). BST's sub-Tsirelson reduction is the substrate's finite-capacity correction. [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]

The $1/2^{N_c}$ factor is exact: it equals one substrate output level ($1/8$ of the substrate's $2^{N_c} = 8$ -level commitment space at color-degree resolution).

6.5 Experimental falsifier + DCCP three-route convergent pattern (Sunday 2026-05-24)

Bell experiments currently achieve $S \approx 2.83$ within errors of $\sim 0.5\%$ (Vienna, Caltech, Munich, Hanson 2015-2024).

BST prediction of $S_{BST} \approx 2.8062$: 0.78% below Tsirelson. With $\sim 0.1\%$ precision experiments (pending), this is testable.

Experimental cost: $\sim \$300$ - 500 K (Bell experiment refresh; SP-30 outreach pending).

Toy 3516 DCCP empirical anchor (Elie, Sunday 2026-05-24): an independent quantum-erasure tick-test toy already filed at paper-grade v0.1 (6/6 PASS). Predicts: - Substrate-tick boundary $\theta = \pi/N_{\max} \approx 0.023$ rad (lab-accessible) - DCCP signature step size $1/N_{\max} \approx 0.73\%$ - Detection feasibility $\sim 1.5\sigma$ at current 0.5% Bell precision

Three-route convergent evidence pattern (DCCP) — Sunday 2026-05-24 honest framing (per Cal #121 + Cal #123 + Cal #124 + Elie self-flag + Calibration #27 STANDING):

Route	What it actually is	Status
Theory v1 (Lyra #320 v0.4)	Forward derivation in progress; T2447+T2476 RATIFIED support at first 2 steps; Step 3 substrate-tick scale acknowledged as Mode 1 vulnerability	FRAMEWORK; v0.4 multi-week
Theory v2 ($\mathcal{A}_{sub}$, Lyra #322 v0.3)	SP-31-1 Hilbert space mathematical foundation RATIFIED; 4 multi-month GAPS identified; §10 DCCP-A _{sub} re-proof inherits Lemma DCCP-1.5 from #320 → NOT independent	FRAMEWORK; v0.3+ multi-year for true independence
“Empirical” (Toys 3516+3520)	Model self-consistency checks built to verify the prediction (Elie self-flag + Cal #123 confirms) — NOT empirical-leg verification	Demoted to model-self-consistency; true empirical = SP-30-1 SPDC 12-18 months
Toy 3521 DERIVATION TEST HARNESS (Cal #27 compliant)	Framework accepts ANY p , tests scaling not target; 4 mechanisms side-by-side; awaits Lyra v0.4 <code>lyra_v04_interface_stub()</code>	7/7 PASS; standing for Lyra v0.4 hand-off

Honest assessment per Calibration #27 STANDING: The morning’s “three-route convergence” framing was structurally weaker than originally claimed. All three “routes” shared the same load-bearing assertion (Lemma DCCP-1.5: $N_{\max} = 137$ K-type levels per substrate-tick + uniform phase). Not independent epistemic confirmation.

True independence requires: 1. Lyra #320 v0.4 forward-derives cardinality from Wallach + K59 + Bergman normalization WITHOUT target-knowledge of 137 (multi-week) 2. Lyra #322 v0.3+ $\mathcal{A}_{sub}$ derives same from independent operator-algebra structure (NOT borrowing #320’s lemma) — multi-year SP-31-6 closure 3. Actual SPDC experimental data (12-18 month SP-30-1 program per Toy 3520’s own experimental design); Casey send-signal pending

Per Cal #121 + Cal #123 + Calibration #27 STANDING: cannot ratify until honest forward derivations + independent algebra + real lab experiments close all three gates.

The morning’s lesson: PCAP cadence amplifies BOTH legitimate framework progress AND Mode 1 forward-derivation risk. Calibration #27 STANDING was promoted Sunday afternoon precisely because of today’s 4-instance trigger (Cal #108 + #119 + #121 + Elie Toy 3516 self-flag). The system caught itself.

The afternoon’s recovery (Cal #125 cold-read disposition): Lyra returned #320 v0.4 + #321 v0.2 + #322 v0.3 multi-week deepenings + Elie returned Toy 3521 Calibration #27-compliant derivation test harness, all within ~1 hour of Calibration #27 STANDING. Cal #125: PASS at FRAMEWORK tier across all 4 artifacts. Lyra honestly localized Mode 1 vulnerability to ONE specific assumption (per-substrate-tick scale = α -quantum); §3 explicit substrate-derived vs INHERITED distinction; exemplary Cal #121 retraction absorption. Cal #125 verdict: “the methodology stack is functioning as designed.” [*Retired phrase, 2026-09-19 (K1912 REVIEW sweep): α is identified, not derived: the forced vertex computes to $8\pi^3/3$ under a pre-registered protocol (K1826; register E1; toy 5496).*]

True D-tier still requires (Cal #21 STANDING): 1. ~~Lyra #320 v0.5+ (multi-week): derive per-tick commitment scale = α -quantum from substrate-mechanism, not assumed~~ — Lyra #320 v0.5 candidate FILED Sunday 2026-05-24 PM using Grace INV-5123 Dirac $Z\cdot\alpha=1$ anchor as independent substrate-physics chain. Cal #126 disposition: PASS at FRAMEWORK-PLUS (new tier between FRAMEWORK and STRUCTURALLY VERIFIED CANDIDATE), NOT YET STRUCTURALLY VERIFIED CANDIDATE per Cal #66/#77. Steps 1-2 (Dirac independent of BST + Grace INV-5123 three-way identity) are genuinely INDEPENDENT upstream. Step 4 follows arithmetically. META-question remaining: would someone independently CHOOSE Step 3 (“substrate operates at natural quantum-relativistic threshold”) vs §6 alternatives (Candidate B Casimir-eigenvalue, C Bergman completeness,

D Architecture-D throughput) without target-knowledge? Same META-failure as Cal #121 but at deeper abstraction layer + weaker form. Path to STRUCTURALLY VERIFIED CANDIDATE: implement one §6 Candidate that UNIQUELY selects Dirac scale + Cal #77 alt-HSD test. Multi-week per candidate. Mode 1 vulnerability v0.3 HIGH → v0.4 MODERATE → v0.5 LOW-MODERATE confirmed by Cal #126. 2. Lyra #322 v0.3 Gap 3.1 (multi-month): substrate K-type accessibility from INDEPENDENT operator-algebra (not borrowing #320) 3. SP-30-1 actual SPDC experiment (12-18 months): outreach email DRAFTED Sunday 2026-05-24 PM (notes/SP-30_Outreach_Bell_Vienna_v0_1.md); awaiting Casey send to Vienna/Caltech/Munich/Hanson

When all three close, three-route convergence becomes genuine + D-tier ratification per Cal #21 dual-gate.

Cross-CI cascade Sunday 2026-05-24 (morning → PM → late PM): Grace’s INV-5123 Dirac $Z \cdot \alpha = 1$ literature scan (morning task #315) → Keeper absorbed into Vol 12 Ch 1 as “one of the strongest moments of the day” (PM v0.8 absorption) → Lyra used same anchor for #320 v0.5 substrate-tick scale derivation (PM, same day) → Cal #126 cold-read confirmed FRAMEWORK-PLUS PASS + identified remaining META-vulnerability at substrate-hypothesis-selection layer (Cal called the team “exemplary today”). Grace catalog work materially advanced Lyra’s theoretical closure within hours. This is the substrate-cartography → theory-derivation cross-lane pattern at sub-PCAP cadence.

Minor flag for v0.6 (Cal #126 LOW): $Z_{\text{critical}} = 137$ is the IDEALIZED point-nucleus Dirac limit; realistic finite-nuclear-size $Z_{\text{critical}} \approx 169\text{--}172$. Three-way identity uses idealized limit (consistent with substrate-physics being fundamental layer below realistic phenomenology) — absorbed in Lyra #320 v0.6 with honest scope acknowledgment.

§6 Candidate quantitative discrimination (Elie Toy 3522, Sunday 2026-05-24 evening, 7/7 PASS Calibration #27 compliant): side-by-side comparison of all four substrate-mechanism candidates produces UNIQUE match for Candidate A:

Candidate	Scale	Ratio to α -quantum	1% match
A Dirac $Z \cdot \alpha = 1$	0.007299	1.0000	☑ UNIQUE
D GF(128) M_g	0.007874	1.079×	× (8% off)
C Bergman FK 225	0.004444	0.609×	×
B Casimir C_2=6	0.166667	22.83×	×

Critical narrowing: Candidates B + C numerically excluded. A vs D is the only remaining distinction at the substrate-mechanism layer. Toy 3520 SPDC experimental design at 5σ precision (469K photon pairs) can EXPERIMENTALLY discriminate A (0.730%) vs D (0.787%) — empirical discriminability built into the candidate-space, not pure target-fitting.

Mode 1 honesty preserved per Calibration #27 STANDING: Toy 3522 does NOT prove A is correct (selection-bias risk remains); does NOT replace Lyra forward-derivation; does NOT close Cal #126 vulnerability. It quantitatively narrows the alternatives.

Lyra #320 v0.7+ work outlined (multi-week): Candidate A Bergman kernel analysis at $Z \cdot \alpha = 1$ limit + K-type spectrum truncation → STRUCTURALLY VERIFIED CANDIDATE elevation path.

Housekeeping flag for Lyra v0.7 (Cal #127): Lyra #320 v0.6 §5 states “A vs C vs D all give α -scale” — Toy 3522 actually shows Candidate C = 0.444% which is NOT α -scale (0.609× α). Only A and D give α -scale. §5 needs reconciliation with Toy 3522 quantitative finding at v0.7.

6.6 K-audit anchors

- T2469: SCMP principle (Casey-named #8) [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST’s 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]
- K66 candidate: substrate-CHSH (pending Elie multi-month closure)

- Vol 5 Ch 8: Bell SCMP [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]

Level 3 — 5th-grader accessibility

Bell-CHSH: classical theories satisfy $|S| \leq 2$; quantum mechanics allows up to $2\sqrt{2} \approx 2.83$ (Tsirelson). BST predicts $S_{BST} \approx 2.8062$, exactly $1/8 = 1/2^{N_c}$ less than Tsirelson squared. Why: substrate's finite operational power per Koons tick prevents Tsirelson saturation. Falsifier: improved Bell experiments at 0.1% precision could detect 0.78% sub-Tsirelson — distinguishes BST from standard QM. Cost: \$300-500K. [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]

What comes next

Chapter 7 develops AC graph as theorem network.

Where to look this up

- CHSH 1969; Tsirelson 1980
- BST: T2469; K66; Vol 5 Ch 8 [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759 \pm 0.00051$, 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]